

ReSTAR: Reliability-Gated Synthetic Candidate Selection for Targeted Augmentation in Radiographic Findings

Anonymous submission

Abstract

Multi-label chest X-ray (CXR) classification aims to identify multiple radiographic findings from a single CXR image. Existing CXR classification systems have achieved strong performance by learning associations between radiographs and their corresponding labels. Unfortunately, these systems often perform well only for certain categories, resulting in inconsistent classification quality across findings. A major cause of this inconsistency is that the underperforming categories often lie in the long tail of the label distribution. While various augmentation techniques can be considered to mitigate this long-tailed class imbalance, generation-based augmentation using recent generative models is a promising approach due to their ability to synthesize highly realistic radiographs. However, these generated images often fall short of serving as effective training data for actual CXR classification. This limitation arises because generative models exhibit a low frequency of satisfying user intent when generating positive samples with clinical findings, thereby maintaining a persistent reliability gap in their practicality as training data. To address this reliability gap, we present ReSTAR, a reliability-gated framework that screens generated CXR image-prompt pairs before converting them into label-aware augmentation data. ReSTAR filters candidates by radiographic usability, localization support, and cross-model finding evidence, while masking unverified non-target labels. On the MIMIC-CXR and NIH medical benchmarks, our proposed ReSTAR consistently improves performance across various classification baselines with minimal computational overhead (only 1.2 s additional runtime per image). In particular, it achieves an average relative AUPRC improvement of 2.4% across the minor-label evaluations. The code will be made publicly available.

Introduction

Chest radiography is one of the most widely used imaging examinations, and large-scale public datasets have enabled substantial progress in automated chest X-ray (CXR) interpretation (Wang et al. 2017; Irvin et al. 2019; Johnson et al. 2019). A common downstream setting is multi-label CXR finding classification, where a model predicts multiple radiographic findings from a single CXR image. Despite many

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

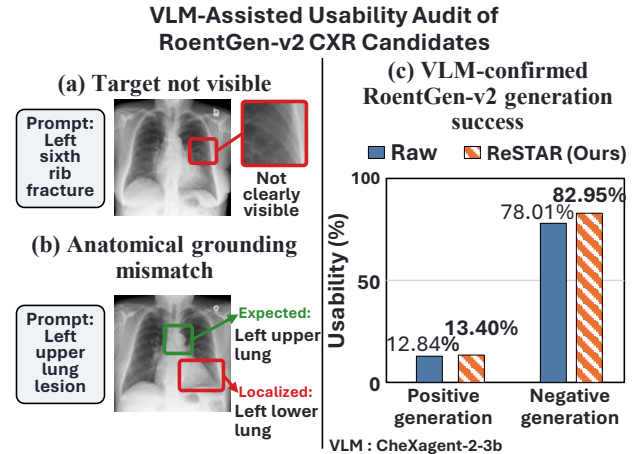


Figure 1: Motivation for reliability-gated candidate selection. Naive generation-based candidates (e.g., RoentGen-v2) are often unreliable due to (a) missing requested findings and (b) incorrectly localizing findings. (c) The VLM audit compares the generation-based and ReSTAR-selected candidates under positive and negative prompt conditions. Usability is a large VLM-assisted proxy for image readability and satisfaction of an intended prompt condition.

x-ray datasets (Holste et al. 2022; Lin et al. 2025) provided to understand CXR images, they still suffer from class imbalance, leading to poor classification performance on CXR images with clinical findings.

Specifically, increased accessibility to CXR imaging and its widespread use in routine screening have led to a massive volume of normal scans (i.e., Negative CXRs), making cases with actual radiographic findings (i.e., Positive CXRs) relatively scarce. Furthermore, even among the positive CXR images, specific images are available in much smaller numbers because their findings (e.g., enlarged cardiomeastinum, consolidation) are extremely rare and scarce, leading to long-tailed distributions and causing severe performance degradation in CXR-based classification systems (Holste et al. 2022; Lin et al. 2025).

An intuitive solution for this imbalance is to enlarge the pool of positive samples, such that standard augmentation including resampling and oversampling can increase label ex-

A. Synthetic Candidate Generation and Admission

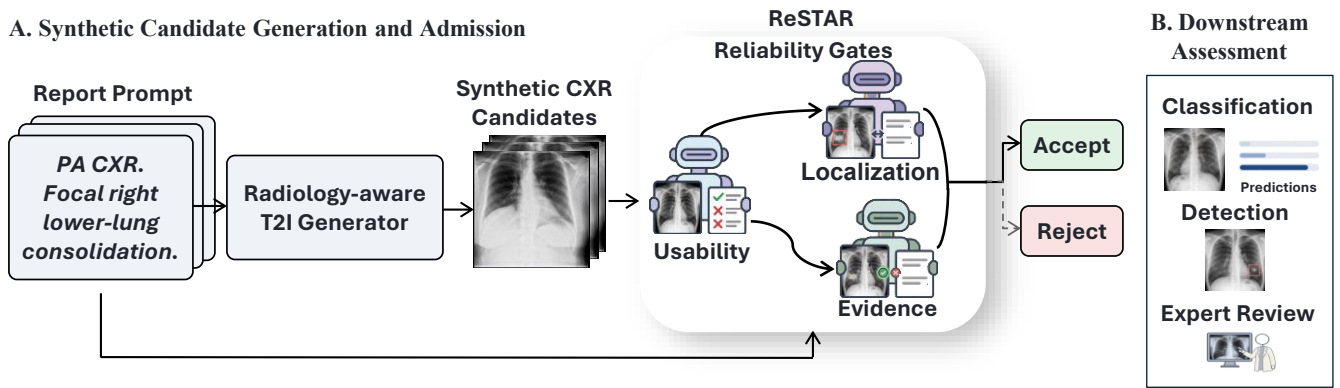


Figure 2: Overview of the ReSTAR framework. (A) Report-style prompts are converted into synthetic CXR candidates by a radiology-aware text-to-image (T2I) generator. Each CXR image–prompt pair is screened by our designed reliability gates referred to as ReSTAR assessing radiographic usability, localization support, and cross-model finding evidence. Candidates satisfying the admission policy are accepted, while the remaining candidates are rejected. (B) The accepted candidates can be used in many downstream assessments (e.g., classification, detection), together with a complementary blinded expert review (human evaluation). Expert review is used only for qualitative assessment.

posure by transforming, duplicating, or reweighting already observed radiographs. These operations are useful for balancing counts, but they cannot create diverse target-positive radiographs, thereby limiting their practical utility in robust clinical deployment. Recent advances in generative modeling (Ho, Jain, and Abbeel 2020; Rombach et al. 2022; Lipman et al. 2023) have enabled the synthesis of realistic and diverse medical images. To be specific, diffusion-based generations (Chambon et al. 2022; Moroianu et al. 2025) offers an creative way to synthesize diverse candidate positive CXR conditioned on textual prompt.

However, generative model-based synthesis of positive-finding CXRs is still not practical enough to serve as effective training data. While a prompt requests the generation of a specific finding, it remains uncertain whether the synthesized image accurately manifests the pathology in an anatomically natural location. Even when it does, the success rate for producing high-fidelity, realistic samples is exceptionally low. Figure 1 illustrates two highly common cases where generative model (Moroianu et al. 2025) takes prompts as input to synthesize positive findings within CXR images, but the resulting image remains unadoptable due to its clinical and anatomical implausibility. Specifically, Figure 1(a) illustrates an instance where the intended pathology is generated but remains visually imperceptible, whereas (b) depicts a failure to accurately localize the pathology despite the prompt specifying the target anatomical position. Quantitatively, when attempting to generate positive and negative CXR images in equal proportions, the number of viable candidates among the generated positive images was substantially lower. Consequently, this leads to a severe scarcity of valid samples available for positive-finding data augmentation. This severe reliability gap between generated positive and negative samples suggests that indiscriminate use of synthesized outputs will only worsen the class imbalance, thereby necessitating a precise framework to selectively identify and retrieve truly meaningful positive examples.

To address this reliability gap, we present ReSTAR, a reliability-gated synthetic candidate selection framework. Rather than treating generated CXRs as training data by default, ReSTAR treats each generated image–prompt pair as a candidate and screens it through complementary reliability gates. These gates assess radiographic usability, localization support and cross-model finding evidence before accepted candidates are converted into conservatively labeled augmentation data. Figure 2 summarizes this validation-first workflow: prompt-conditioned generation expands the candidate pool for underrepresented findings, while reliability gates determine which generated image–prompt pairs are reliable enough to be retained as conservatively labeled augmentation examples. This separation is central to ReSTAR. Generation addresses the quantity side of class imbalance by producing additional rare-finding candidates, whereas reliability-gated selection addresses the trustworthiness side by reducing the gap between increased synthetic counts and usable training signal.

Related Work

Long-Tailed CXR Classification

ChestX-ray14 (Wang et al. 2017), CheXpert (Irvin et al. 2019), and MIMIC-CXR-JPG (Johnson et al. 2019) exhibit severe multi-label imbalance, leaving few positive examples for tail findings. Long-tailed CXR benchmarks (Holste et al. 2022; Lin et al. 2025) evaluate performance across head and tail labels, including zero-shot settings. Prior work improves recognition; ReSTAR instead determines which generated candidates enter its augmentation set. CheXpert (Irvin et al. 2019) and MIMIC-CXR-JPG (Johnson et al. 2019) use report-derived labels, whereas localization-annotated resources, such as VinDr-CXR (Nguyen et al. 2020), remain scarce. Accordingly, increasing tail-label counts is insufficient unless each added image supports its intended finding and unverified labels are handled conservatively.

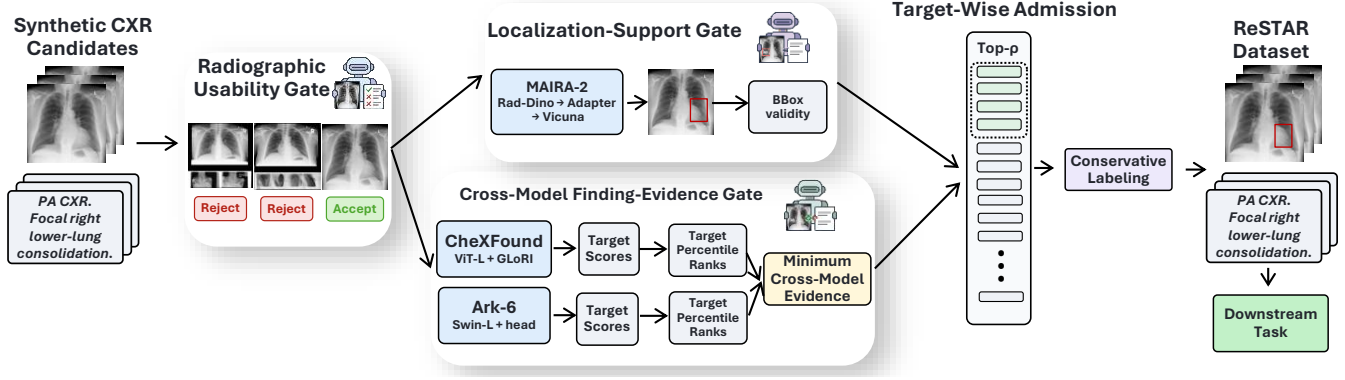


Figure 3: Overview of ReSTAR candidate selection and dataset construction. Synthetic image–prompt pairs pass through the Radiographic Usability Gate, Localization-Support Gate, and Cross-Model Finding-Evidence Gate. Candidates satisfying all target-wise criteria form the ReSTAR set, with prompted findings labeled positive and unprompted findings masked.

Synthetic CXR Generation

Text-conditioned CXR generation increasingly uses radiology-aware diffusion models. RoentGen (Chambon et al. 2022) generates radiographs from free-form text, while RoentGen-v2 (Moroianu et al. 2025) provides finer control over findings and demographic attributes. Chest-Diffusion (Huang et al. 2024a) improves generation efficiency, Diff-CXR (Huang et al. 2024b) incorporates disease-aware conditioning, and CXRL (Han et al. 2024) strengthens prompt following through comparative feedback. Evaluation has also moved beyond visual realism. CheXGenBench (Dutt et al. 2025) assesses fidelity, privacy, and downstream utility, whereas CSEval (Cronshaw et al. 2026) evaluates whether generated images preserve the clinical semantics of their prompts. These advances improve candidate generation and evaluation, but they do not define a training-data admission policy that jointly checks usability, target localization, and finding evidence while accounting for incomplete synthetic labels.

Reliability Screening for Synthetic CXRs

Selective augmentation studies (Xue et al. 2019; Ye et al. 2020; Pattilachan et al. 2022) show that quality assurance and sample selection can improve synthetic medical data. Radiology-specific models provide complementary screening signals: MAIRA-2 (Bannur et al. 2024) grounds reported findings in CXR regions, while CheXFound (Yang et al. 2025) and Ark (Ma et al. 2023) provide separate finding scores. Diffusion memorization studies (Carlini et al. 2023) further motivate similarity auditing before downstream reuse. No individual signal establishes clinical validity; their failure modes also differ. ReSTAR therefore combines radiographic usability, target-specific localization support, and cross-model finding evidence as non-compensatory candidate-level gates. Accepted candidates receive prompted positive labels, while unprompted findings remain masked. Thus, ReSTAR contributes an admission-and-labeling framework for synthetic CXR augmentation rather than a new image generator.

Methods

Synthetic Candidate Construction

Let $\mathcal{C} = \{1, \dots, K\}$ denote CXR finding labels with the number of K for downstream classifier, and let $\mathcal{T}_{\text{syn}} \subseteq \mathcal{C}$ denote the findings targeted for synthesis. The i -th candidate is represented as $u_i = (x_i, p_i, T_i, \Phi_i)$, where $x_i \in [0, 1]^{C_{\text{img}} \times H \times W}$ is a generated CXR, p_i is its report-style prompt, $\emptyset \neq T_i \subseteq \mathcal{T}_{\text{syn}}$ is the prompted target set, and $\Phi_i = \{\phi_{i,t} : t \in T_i\}$ contains the target-specific phrases used for localization. Here, C_{img} , H , and W denote the image channels, height, and width after preprocessing.

Prompts are deterministically constructed from target specifications and label combinations observed in the MIMIC-CXR training split (Johnson et al. 2019). Uncertain findings are not prompted as positives, and target-mention, negation, and *No Finding* conflicts are rejected during construction. Each prompt is passed to RoentGen-v2 (Moroianu et al. 2025) to generate x_i . Each resulting image–prompt pair remains a candidate rather than a verified training example. Subsequently, these candidates pass through our designed gates (i.e., usability, localization, evidence) to select highly effective samples for training in the following.

Reliability-Gated Candidate Selection

ReSTAR applies three complementary gates. The Radiographic Usability Gate removes gross generation failures, while the Localization-Support and Cross-Model Finding-Evidence Gates provide target-specific admission signals. Here, a gate denotes an operational screening component rather than clinical validation.

Radiographic Usability Gate. This gate rejects decoding failures, blank or near-blank images, extreme-intensity images, large internal bands or seams, and separator- or panel-like layouts. Let $a_i \in \{0, 1\}$ denote its output, with $a_i = 1$ indicating a pass. The usability-feasible index set $\mathcal{F}$ contains the indices of candidates with $a_i = 1$ that also satisfy the prompt-integrity constraints. This gate is a deterministic veto for gross failures rather than a learned estimator of diagnostic quality.

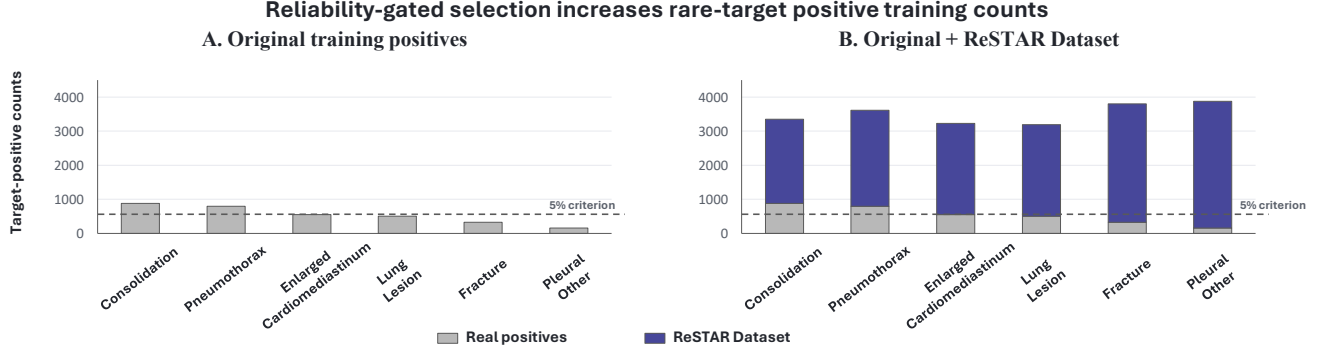


Figure 4: Rare-target positive counts before and after reliability-gated augmentation. Panels A and B share the same y-axis scale; the dashed line marks the 5% low-prevalence criterion, where ReSTAR effectively augments low-prevalence findings.

Localization-Support Gate. For each $i \in \mathcal{F}$ and $t \in T_i$, MAIRA-2 (Bannur et al. 2024) produces target-specific localization boxes. The binary localization signal is defined as: $g_{i,t} = \mathbb{1}[\mathcal{B}_{i,t}^{\text{valid}} \neq \emptyset]$, where $\mathcal{B}_{i,t}^{\text{valid}}$ contains boxes satisfying the predefined parsing and geometric-validity criteria.

Admission requires $g_{i,t} = 1$ for every $t \in T_i$; finding evidence cannot compensate for missing localization. Boxes retained for admitted candidates are treated as pseudo-localization annotations rather than exhaustive or radiologist-verified ground truth.

Cross-Model Finding-Evidence Gate. For each $i \in \mathcal{F}$ and $t \in T_i$, CheXFound (Yang et al. 2025) and Ark-6 (Ma et al. 2023) independently produce finding scores $s_{i,t}^{(m)}$. Let $\mathcal{M} = \{\text{CheXFound}, \text{Ark-6}\}$. Because score scales differ across models and targets, we define the cross-model finding-evidence score $e_{i,t}$ as:

$$e_{i,t} = \min_{m \in \mathcal{M}} \hat{F}_t^{(m)}(s_{i,t}^{(m)}). \quad (1)$$

Here, $\hat{F}_t^{(m)}$ is the empirical CDF of model m for target t , estimated over the corresponding candidates in $\mathcal{F}$. Thus, $e_{i,t} \in [0, 1]$ is the lower of the target-specific percentile scores produced by the two models. A high score from one model therefore cannot compensate for weak support from the other.

Target-Wise Admission. Given a target-wise retention parameter $\rho \in (0, 1]$, the target-specific evidence threshold $\tau_{t,\rho}$ is defined as:

$$\tau_{t,\rho} = Q_{1-\rho}(\{e_{j,t} \mid j \in \mathcal{F}, t \in T_j\}), \quad (2)$$

where Q_q denotes the empirical q -quantile. Thus, $\tau_{t,\rho}$ is the cutoff corresponding approximately to the upper ρ fraction of candidates for target t , before localization and multi-target filtering.

The final admitted candidate index set, denoted by $\mathcal{I}_\rho$, is defined as:

$$\mathcal{I}_\rho = \{i \in \mathcal{F} \mid g_{i,t} = 1 \text{ and } e_{i,t} \geq \tau_{t,\rho}, \forall t \in T_i\}. \quad (3)$$

Equation (3) imposes a non-compensatory rule: every prompted target must satisfy both localization and evidence requirements.

Conservative Labeling

Each admitted candidate is treated as a partially annotated CXR. For $i \in \mathcal{I}_\rho$, the label mask is defined as: $m_{i,\ell} = \mathbb{1}[\ell \in T_i]$ for $\ell \in \mathcal{C}$, with $\hat{y}_{i,\ell} = 1$ for every prompted label. Unprompted labels are treated as unobserved rather than negative.

For a synthetic mini-batch $\mathcal{B}_{\text{syn}}$, let $\pi_{i,\ell}$ denote the predicted probability for label ℓ . The synthetic data loss $\mathcal{L}_{\text{syn}}$ is defined with masked binary cross-entropy (BCE) loss as:

$$\mathcal{L}_{\text{syn}} = \frac{\sum_{i \in \mathcal{B}_{\text{syn}}} \sum_{\ell \in \mathcal{C}} m_{i,\ell} \text{BCE}(\pi_{i,\ell}, \hat{y}_{i,\ell})}{\sum_{i \in \mathcal{B}_{\text{syn}}} \sum_{\ell \in \mathcal{C}} m_{i,\ell}}. \quad (4)$$

Equation (4) averages the BCE only over prompted positive entries; unprompted labels contribute no synthetic loss.

Let $\mathcal{L}_{\text{real}}$ denote the multi-label loss for a real mini-batch. The overall classification loss is defined as: $\mathcal{L}_{\text{cls}} = \mathcal{L}_{\text{real}} + \lambda_{\text{syn}} \mathcal{L}_{\text{syn}}$, where λ_{syn} controls the contribution of synthetic examples.

ReSTAR Dataset

We define the *ReSTAR dataset* as synthetic candidates indexed by $\mathcal{I}_\rho$, together with their report-style prompts, prompted positive-label masks, and valid target-specific MAIRA-2 boxes. Unprompted findings remain masked, while the boxes serve as pseudo-localization annotations.

Figure 4 compares the original positive counts with those added by ReSTAR for six low-prevalence findings. The dashed line marks the 5% selection criterion in the original MIMIC-CXR training split. These synthetic counts represent added positive annotations rather than post-augmentation prevalence.

Experiments

Experiment Setup

We generate 20,000 CXR image-prompt candidates using RoentGen-v2. The Radiographic Usability Gate retains 19,530 candidates, and target-wise admission with $\rho = 0.4$

Type	Backbone	MIMIC-CXR		NIH-overlap	
		AUPRC	AUROC	AUPRC	AUROC
Full	DenseNet-121	0.3064±0.0016	0.7516±0.0025	0.1573±0.0033	0.7207±0.0049
	DenseNet-121 + ReSTAR	0.3078±0.0022	0.7516±0.0029	0.1618±0.0037	0.7259±0.0044
	ResNet-18	0.3079±0.0014	0.7496±0.0017	0.1633±0.0032	0.7278±0.0062
	ResNet-18 + ReSTAR	0.3136±0.0025	0.7503±0.0021	0.1649±0.0034	0.7284±0.0070
	ConvNeXt-Tiny	0.3265±0.0020	0.7735±0.0022	0.1762±0.0041	0.7506±0.0015
	ConvNeXt-Tiny + ReSTAR	0.3276±0.0016	0.7739±0.0008	0.1764±0.0039	0.7505±0.0032
Minor	DenseNet-121	0.0760±0.0030	0.6973±0.0067	0.0661±0.0044	0.7254±0.0102
	DenseNet-121 + ReSTAR	0.0776±0.0033	0.6991±0.0073	0.0693±0.0083	0.7334±0.0105
	ResNet-18	0.0773±0.0010	0.6956±0.0025	0.0688±0.0060	0.7265±0.0138
	ResNet-18 + ReSTAR	0.0781±0.0019	0.6980±0.0028	0.0721±0.0068	0.7383±0.0164
	ConvNeXt-Tiny	0.0910±0.0023	0.7297±0.0044	0.0927±0.0050	0.7629±0.0016
	ConvNeXt-Tiny + ReSTAR	0.0918±0.0020	0.7308±0.0030	0.0934±0.0086	0.7635±0.0072

Table 1: Effectiveness analysis with and without ReSTAR on MIMIC-CXR and NIH-overlap using three different classification backbones. AUPRC and AUROC values are reported with 95% confidence intervals (mean $\pm 1.96 \times$ standard error) over five random seeds. Full and Minor denote the full and low-prevalence label sets, respectively.

yields 14,065 ReSTAR samples for classification. For detection, ReSTAR contains 1,200 synthetic images and 1,248 MAIRA-2-derived boxes, while the size-matched RoentGen-based baseline contains 1,185 images and 1,240 boxes.

All experiments are repeated using five random seeds. Classification models are trained for 12 epochs using AdamW with a learning rate of 1×10^{-4} and a weight decay of 1×10^{-4} . Each classification update uses a real mini-batch $\mathcal{B}_{\text{real}}$, with $|\mathcal{B}_{\text{real}}| = 32$. In the synthetic-augmentation settings, an additional synthetic mini-batch $\mathcal{B}_{\text{syn}}$ is used, with $|\mathcal{B}_{\text{syn}}| = 8$. We set $\lambda_{\text{syn}} = 1$ for all classification experiments using synthetic augmentation. Detection models are trained for 10 epochs with a batch size of 2 using AdamW (Loshchilov and Hutter 2019) with a learning rate of 1×10^{-4} and a weight decay of 1×10^{-4} . All experiments are conducted on NVIDIA Quadro RTX 8000 GPUs with 48 GB of memory.

Baselines. For cross-backbone classification, DenseNet-121 (Huang et al. 2017), ResNet-18 (He et al. 2016), and ConvNeXt-Tiny (Liu et al. 2022) are each trained under two data settings. All backbones are initialized with ImageNet-pretrained weights (Deng et al. 2009). We conducted the ablation study on DenseNet-121, as it is widely established as the standard backbone in this domain.

For detection, we compare *Real only*, *RoentGen-based 1.2K*, and *ReSTAR 1.2K* using Faster R-CNN (Ren et al. 2015) with a ResNet-50 (He et al. 2016) backbone and a Feature Pyramid Network (Lin et al. 2017), initialized from COCO-pretrained weights (Lin et al. 2014).

Dataset. For classification, we utilize a subset of the MIMIC-CXR dataset (Johnson et al. 2019), which consists of 18,000 training, 1,999 validation, and 10,000 test images. The classifier is trained to predict all 14 MIMIC-CXR labels. To perform out-of-domain evaluation, the NIH ChestX-ray14 dataset (Wang et al. 2017) serves as an additional test set. For this evaluation, we focus on the eight labels shared with MIMIC-CXR (i.e., No Finding, Effusion, Atelectasis, Cardiomegaly, Edema, Pneumonia, Consolidation, and Pneumothorax), which we refer to as NIH-overlap.

To evaluate performance on the tail of the label distribution, we define the six MIMIC-CXR findings with training prevalence below 5% as the *minor label set* (i.e., Consolidation, Pneumothorax, Enlarged Cardiomediastinum, Lung Lesion, Fracture, and Pleural Other), representing the tail of the label distribution. Thus, classification results are reported for both the full label set and the minor label set. The corresponding NIH-overlap minor label set includes Consolidation and Pneumothorax, the two minor labels shared with MIMIC-CXR.

For detection, we use the six minor findings and the real VinDr-CXR dataset (Nguyen et al. 2020). The training and validation splits contain 3,128 images with 6,558 target boxes and 602 images with 281 target boxes, respectively. Classification images are resized to 224×224 and normalized using ImageNet statistics. Training augmentation consists of random affine transformations with rotations up to $\pm 7^\circ$, translations up to 3%, and scaling in $[0.95, 1.05]$, followed by brightness and contrast jitter of 0.08. Detection images are resized to a minimum side of 800 pixels and a maximum side of 1,333 pixels, without additional augmentation.

Evaluation Metrics. For classification, we report macro AUPRC as the primary metric and macro AUROC as a complementary metric. Both are computed as the unweighted mean of per-finding scores. For MIMIC-CXR, we average over all 14 findings for the full label set and the six low-prevalence findings for the minor label set. For NIH-overlap, the corresponding averages cover the eight shared findings and the two shared minor findings, Consolidation and Pneumothorax.

For detection, we report six-target macro AP50 at an IoU threshold of 0.50. We additionally report the mean AP50 for Pneumothorax and Fracture, the two sparsest findings in the 602-image validation set, with 5 and 6 ground-truth boxes, respectively.

Experiment Results

Cross-backbone classification. Table 1 evaluates ReSTAR across three classification backbones. Each

Type	Method	N_{syn}	MIMIC-CXR		NIH-overlap	
			AUPRC	AUROC	AUPRC	AUROC
Full	Real only	0	0.3064	0.7516	0.1573	0.7207
	RoentGen	14,065	0.3055	0.7563	0.1571	0.7159
	ReSTAR	14,065	0.3078	0.7516	0.1618	0.7259
	w/o Usability	14,403	0.3063	0.7492	0.1579	0.7110
	w/o Localiz.	14,640	0.3057	0.7522	0.1590	0.7251
	w/o Evidence	18,726	0.3077	0.7544	0.1591	0.7226
Minor	Real only	0	0.0760	0.6973	0.0661	0.7254
	RoentGen	14,065	0.0742	0.7075	0.0693	0.7258
	ReSTAR	14,065	0.0776	0.6991	0.0693	0.7334
	w/o Usability	14,403	0.0755	0.6946	0.0591	0.7001
	w/o Localiz.	14,640	0.0749	0.6985	0.0677	0.7324
	w/o Evidence	18,726	0.0771	0.7031	0.0665	0.7244

Table 2: Classification performance and ablation of the ReSTAR reliability gates on the full and low-prevalence label sets. N_{syn} denotes the number of added synthetic samples. RoentGen denotes unfiltered synthetic augmentation using the same number of samples as ReSTAR.

backbone is trained with real data only and with the same 14,065-sample ReSTAR set. ReSTAR improves mean AUPRC in all 12 backbone-evaluation combinations. AUROC improves in 10 comparisons, remains unchanged for DenseNet-121 on full MIMIC-CXR, and decreases by only 0.0001 for ConvNeXt-Tiny on full NIH-overlap. The largest AUROC gains occur on the NIH-overlap minor-label evaluation: 0.7265 to 0.7383 for ResNet-18 and 0.7254 to 0.7334 for DenseNet-121. Gains on MIMIC-CXR, particularly in AUPRC, are smaller. Overall, the consistent direction across ResNet, ConvNeXt, and DenseNet suggests that ReSTAR is not tied to a specific backbone, although its benefits remain modest and vary across architectures and evaluation distributions.

Ablation study. Using a DenseNet-121 (Huang et al. 2017) classifier initialized with ImageNet-pretrained weights (Deng et al. 2009), Table 2 compares *Real only*, count-matched *RoentGen-based* augmentation, *ReSTAR*, and three leave-one-gate-out variants. RoentGen-based uses 14,065 ungated candidates from the same RoentGen-v2 pool, while each ablation set is reconstructed after removing the corresponding gate and therefore contains a different number of samples. At the matched count, ReSTAR outperforms RoentGen-based in mean AUPRC on the MIMIC-CXR full (0.3078 vs. 0.3055) and minor (0.0776 vs. 0.0742) sets and the NIH-overlap full set (0.1618 vs. 0.1571), while tying on the NIH-overlap minor set (0.0693). ReSTAR achieves the highest NIH-overlap AUROC, whereas RoentGen-based performs best on MIMIC-CXR AUROC, showing that the gains vary by dataset and metric. Removing any gate lowers MIMIC-CXR minor-label and NIH-overlap full-label AUPRC. Notably, *w/o Evidence* retains more samples than ReSTAR (18,726 vs. 14,065) but performs slightly worse on MIMIC-CXR full-label (0.3077 vs. 0.3078) and minor-label AUPRC (0.0771 vs. 0.0776), indicating that retaining more samples does not necessarily improve classification.

Method	Macro AP50	PTX/Fracture AP50
Real only	0.1688 $\pm$ 0.0206	0.0554 $\pm$ 0.0571
RoentGen-based	0.1773 $\pm$ 0.0085	0.0704 $\pm$ 0.0334
ReSTAR (Ours)	0.1786$\pm$0.0098	0.0831$\pm$0.0416

Table 3: Faster R-CNN detection performance on real VinDr-CXR validation images. Macro AP50 is averaged over the six target findings, while PTX/Fracture AP50 is averaged over Pneumothorax and Fracture, the two sparsest findings in the validation set, reported with 95% confidence intervals (mean $\pm 1.96 \times$ standard error) over five random seeds.

VinDr-CXR detection. Table 3 compares Faster R-CNN detectors on the same real VinDr-CXR validation images, which are not used for training. Under approximately matched 1.2K synthetic augmentation, ReSTAR yields the highest mean six-finding macro AP50 (0.1786), compared with real-only training (0.1688) and RoentGen-based augmentation (0.1773). ReSTAR also yields the highest mean AP50 over Pneumothorax and Fracture (0.0831), the two sparsest findings in the validation set, compared with 0.0554 and 0.0704, respectively. However, the margin over RoentGen-based augmentation is modest and the seed variability is substantial, particularly for Pneumothorax and Fracture; we therefore interpret these results as complementary evidence of localization utility rather than a robust detection improvement.

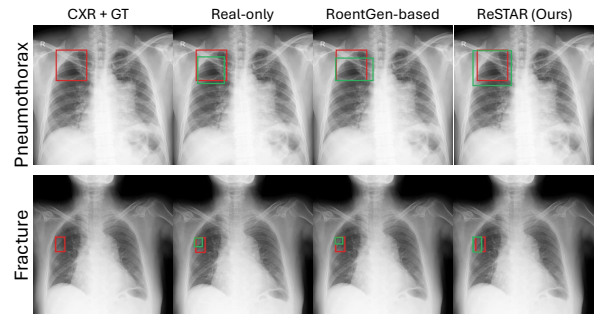


Figure 5: Qualitative VinDr-CXR detection examples for Pneumothorax and Fracture. Red and green boxes denote ground truth and predictions, respectively.

Qualitative results on CXR image detection. Figure 5 presents real VinDr-CXR evaluation images, covering Pneumothorax and Fracture, the two sparsest findings. In each row, the first column shows the ground-truth annotation, while the remaining columns show predictions on the same CXR from detectors trained with (1) real data only, (2) real data augmented with unfiltered synthetic candidates, and (3) real data augmented with the ReSTAR accepted set. In both examples, the ReSTAR prediction overlaps the ground-truth region more closely than the real-only and unfiltered-synthetic predictions, suggesting that the selected synthetic images help the detector localize rare findings more accurately.

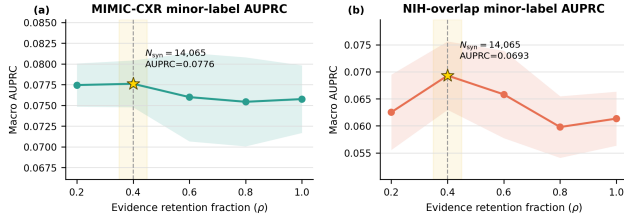


Figure 6: Sensitivity of minor-label macro AUPRC to the per-label retention fraction ρ used by the Cross-Model Finding-Evidence Gate. Lines and shaded regions show the mean and $\pm$ one standard deviation over five seeds, respectively. Stars indicate the setting used in the main experiments ($\rho = 0.4$), which yields $N_{\text{syn}} = 14,065$.

Configuration	Time (sec)
DenseNet-121	32.50
DenseNet-121 + LS (MAIRA-2)	33.67
DenseNet-121 + FE (CheXFound)	32.70
DenseNet-121 + FE (Ark-6)	33.00
DenseNet-121 + ReSTAR (Total)	33.74

Table 4: Average runtime per candidate image for the model-based reliability gates. LS denotes Localization-Support gate and FE denotes Finding-Evidence gate. Total denotes the combination of all three components.

Sensitivity Analysis Figure 6 shows that minor-label macro AUPRC remains broadly stable across the evaluated values of ρ . Although NIH-overlap varies more than MIMIC-CXR, $\rho = 0.4$ yields competitive mean performance on both datasets and is used as a practical common setting ($N_{\text{syn}} = 14,065$). The overlapping standard-deviation bands support robustness to moderate changes in ρ , but do not establish $\rho = 0.4$ as a uniquely optimal setting.

Efficiency Analysis As shown in Table 4, adding the complete ReSTAR screening pipeline increased the measured runtime from 32.50 to 33.74 seconds, corresponding to an overhead of 1.24 seconds, or 3.8%, relative to DenseNet-121 inference alone. Most of this additional cost was attributable to the Localization-Support Gate, while the two Finding-Evidence models contributed only minor increases. Thus, ReSTAR provides candidate-level reliability screening with a relatively small computational overhead compared with the downstream DenseNet-121 inference workload.

Expert review findings (Human Evaluation). Figure 7 shows generated CXR examples and representative post-hoc reviewer notes. The primary reviewer rated 27/30 cases at image quality ≥ 4 , 28/28 evaluable cases as target-visible, and 28/28 evaluable overlays as correctly localized, while major artifacts were still flagged in 15/30 cases. A conservative second review marked target visibility in only 6/30 cases, indicating reviewer-dependent assessment of subtle synthetic findings. We interpret this review as qualitative support for candidate-level reliability assessment.

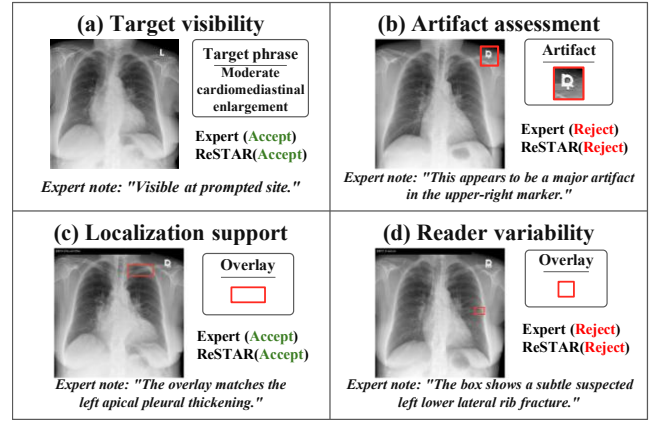


Figure 7: Representative cases from the blinded expert review. Each panel shows a generated CXR image or localization overlay together with a post-hoc reviewer note. Reviewers were blinded to validation-group assignments and gate outputs.

Discussion and Limitations

ReSTAR treats long-tailed CXR augmentation as candidate generation and screening. Because visual plausibility and coarse prompt agreement do not ensure that the target finding appears at the intended anatomical location, ReSTAR assesses radiographic usability, localization support, and cross-model finding evidence before retaining a candidate. In our implementation, screening increased average VRAM by approximately 7.6 GB, which remains manageable relative to GPU memory available for large-model workloads. Because screening occurs offline during dataset construction, the downstream architecture and inference remain unchanged. The gates automate screening but do not replace radiologist validation; retained images may contain artifacts or fail to depict the prompted finding clearly. Our evaluation covered one generator under one generation setting, six low-prevalence findings, eight labels shared by MIMIC-CXR and NIH, and VinDr-CXR detection data with few positive annotations for the rarest findings. The gates may propagate component-model errors, and the fixed per-label retention fraction was not radiologist-calibrated. Limited expert review cannot establish diagnostic validity, and our current screening does not rule out memorization or privacy leakage. Future work should evaluate additional generators, settings, and institutions, calibrate the gates through larger radiologist studies, and conduct formal privacy audits.

Conclusion

We introduced ReSTAR, a reliability-gated framework that screens generated CXR image-prompt candidates for long-tailed augmentation. Classification, gate-ablation, and detection results support screening for radiographic usability, anatomical localization, and finding evidence before training. ReSTAR provides a practical augmentation strategy with manageable additional memory and no changes to downstream model deployment.

References

- Bannur, S.; Bouzid, K.; Castro, D. C.; Schwaighofer, A.; Thieme, A.; Bond-Taylor, S.; Ilse, M.; Pérez-García, F.; Salvatelli, V.; Sharma, H.; Meissen, F.; Ranjit, M.; Srivastav, S.; Gong, J.; Codella, N. C. F.; Falck, F.; Oktay, O.; Lungren, M. P.; Wetscherek, M. T.; Alvarez-Valle, J.; and Hyland, S. L. 2024. MAIRA-2: Grounded Radiology Report Generation. *arXiv preprint arXiv:2406.04449*.
- Carlini, N.; Hayes, J.; Nasr, M.; Jagielski, M.; Schwag, V.; Tramèr, F.; Balle, B.; Ippolito, D.; and Wallace, E. 2023. Extracting Training Data from Diffusion Models. In *32nd USENIX Security Symposium*, 5253–5270.
- Chambon, P.; Bluethgen, C.; Delbrouck, J.-B.; Van der Sluijs, R.; Połacin, M.; Zambrano Chaves, J. M.; Abraham, T. M.; Purohit, S.; Langlotz, C. P.; and Chaudhari, A. 2022. RoentGen: Vision-Language Foundation Model for Chest X-Ray Generation. *arXiv preprint arXiv:2211.12737*.
- Cronshaw, R.; Vilouras, K.; Yan, J.; Du, Y.; Chen, F.; McDonagh, S.; and Tsaftaris, S. A. 2026. CSEval: A Framework for Evaluating Clinical Semantics in Text-to-Image Generation. *arXiv preprint arXiv:2602.12004*.
- Deng, J.; Dong, W.; Socher, R.; Li, L.-J.; Li, K.; and Fei-Fei, L. 2009. ImageNet: A Large-Scale Hierarchical Image Database. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 248–255.
- Dutt, R.; Sanchez, P.; Yao, Y.; McDonagh, S.; Tsaftaris, S. A.; and Hospedales, T. 2025. CheXGenBench: A Unified Benchmark for Fidelity, Privacy and Utility of Synthetic Chest Radiographs. *arXiv preprint arXiv:2505.10496*.
- Han, W.; Kim, C.; Ju, D.; Shim, Y.; and Hwang, S. J. 2024. Advancing Text-Driven Chest X-Ray Generation with Policy-Based Reinforcement Learning. In *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*.
- He, K.; Zhang, X.; Ren, S.; and Sun, J. 2016. Deep Residual Learning for Image Recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 770–778.
- Ho, J.; Jain, A.; and Abbeel, P. 2020. Denoising Diffusion Probabilistic Models. In *Advances in Neural Information Processing Systems*, volume 33, 6840–6851.
- Holste, G.; Wang, S.; Jiang, Z.; Shen, T. C.; Shih, G.; Summers, R. M.; Peng, Y.; and Wang, Z. 2022. Long-Tailed Classification of Thorax Diseases on Chest X-Ray: A New Benchmark Study. *arXiv preprint arXiv:2208.13365*.
- Huang, G.; Liu, Z.; Van Der Maaten, L.; and Weinberger, K. Q. 2017. Densely Connected Convolutional Networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 4700–4708.
- Huang, P.; Gao, X.; Huang, L.; Jiao, J.; Li, X.; Wang, Y.; and Guo, Y. 2024a. Chest-Diffusion: A Light-Weight Text-to-Image Model for Report-to-CXR Generation. *arXiv preprint arXiv:2407.00752*.
- Huang, P.; Guo, B.; Liang, S.; Fu, J.; Wang, Y.; and Guo, Y. 2024b. Diff-CXR: Report-to-CXR Generation Through a Disease-Knowledge Enhanced Diffusion Model. *arXiv preprint arXiv:2410.20165*.
- Irvin, J.; Rajpurkar, P.; Ko, M.; Yu, Y.; Ciurea-Ilcus, S.; Chute, C.; Marklund, H.; Haghighi, B.; Ball, R.; Shpan-skaya, K.; Seekins, J.; Mong, D. A.; Halabi, S. S.; Sandberg, J. K.; Jones, R.; Larson, D. B.; Langlotz, C. P.; Patel, B. N.; Lungren, M. P.; and Ng, A. Y. 2019. CheXpert: A Large Chest Radiograph Dataset with Uncertainty Labels and Expert Comparison. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, 590–597.
- Johnson, A. E. W.; Pollard, T. J.; Greenbaum, N. R.; Lungren, M. P.; Deng, C.-y.; Peng, Y.; Lu, Z.; Mark, R. G.; Berkowitz, S. J.; and Horng, S. 2019. MIMIC-CXR-JPG, a Large Publicly Available Database of Labeled Chest Radiographs. *arXiv preprint arXiv:1901.07042*.
- Lin, M.; Holste, G.; Wang, S.; Zhou, Y.; Wei, Y.; Banerjee, I.; Chen, P.; Dai, T.; Du, Y.; Dvornek, N. C.; Ge, Y.; Guo, Z.; Hanaoka, S.; Kim, D.; Messina, P.; Lu, Y.; Parra, D.; Son, D.; Soto, Á.; Urooj, A.; Vidal, R.; Yamagishi, Y.; Yang, Z.; Zhang, R.; Zhou, Y.; Celi, L. A.; Summers, R. M.; Lu, Z.; Chen, H.; Flanders, A.; Shih, G.; Wang, Z.; and Peng, Y. 2025. CXR-LT 2024: A MICCAI Challenge on Long-Tailed, Multi-Label, and Zero-Shot Disease Classification from Chest X-Ray. *arXiv preprint arXiv:2506.07984*.
- Lin, T.-Y.; Dollár, P.; Girshick, R.; He, K.; Hariharan, B.; and Belongie, S. 2017. Feature Pyramid Networks for Object Detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2117–2125.
- Lin, T.-Y.; Maire, M.; Belongie, S.; Hays, J.; Perona, P.; Ramanan, D.; Dollár, P.; and Zitnick, C. L. 2014. Microsoft COCO: Common Objects in Context. In *European Conference on Computer Vision*, 740–755. Springer.
- Lipman, Y.; Chen, R. T. Q.; Ben-Hamu, H.; Nickel, M.; and Le, M. 2023. Flow Matching for Generative Modeling. In *International Conference on Learning Representations*.
- Liu, Z.; Mao, H.; Wu, C.-Y.; Feichtenhofer, C.; Darrell, T.; and Xie, S. 2022. A ConvNet for the 2020s. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 11976–11986.
- Loshchilov, I.; and Hutter, F. 2019. Decoupled Weight Decay Regularization. In *International Conference on Learning Representations*.
- Ma, D.; Pang, J.; Gotway, M. B.; and Liang, J. 2023. Foundation Ark: Accruing and Reusing Knowledge for Superior and Robust Performance. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2023*, volume 14220, 651–662. Springer Nature Switzerland.
- Moroianu, S. L.; Bluethgen, C.; Chambon, P.; Cherti, M.; Delbrouck, J.-B.; Paschali, M.; Price, B.; Gichoya, J.; Jitsev, J.; Langlotz, C. P.; and Chaudhari, A. S. 2025. Improving Performance, Robustness, and Fairness of Radiographic AI Models with Finely-Controllable Synthetic Data. *arXiv preprint arXiv:2508.16783*.
- Nguyen, H. Q.; Lam, K.; Le, L. T.; Pham, H. H.; Tran, D. Q.; Nguyen, D. B.; Le, D. D.; Pham, C. M.; Tong, H. T. T.; Dinh, D. H.; Do, C. D.; Doan, L. T.; Nguyen, C. N.; Nguyen, B. T.; Nguyen, Q. V.; Hoang, A. D.; Phan, H. N.; Nguyen, A. T.; Ho, P. H.; Ngo, D. T.; Nguyen, N. T.; Nguyen, N. T.; Dao, M.; and Vu, V. 2020. VinDr-CXR: An Open Dataset of

Chest X-Rays with Radiologist’s Annotations. *arXiv preprint arXiv:2012.15029*.

Patilachan, T. M.; Demir, U.; Keles, E.; Jha, D.; Klatte, D.; Engels, M.; Hoogenboom, S.; Bolan, C.; Wallace, M.; and Bagci, U. 2022. A Critical Appraisal of Data Augmentation Methods for Imaging-Based Medical Diagnosis Applications. *arXiv preprint arXiv:2301.02181*.

Ren, S.; He, K.; Girshick, R.; and Sun, J. 2015. Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks. In *Advances in Neural Information Processing Systems*, volume 28.

Rombach, R.; Blattmann, A.; Lorenz, D.; Esser, P.; and Ommer, B. 2022. High-Resolution Image Synthesis with Latent Diffusion Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 10684–10695.

Wang, X.; Peng, Y.; Lu, L.; Lu, Z.; Bagheri, M.; and Summers, R. M. 2017. ChestX-ray8: Hospital-Scale Chest X-Ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2097–2106.

Xue, Y.; Ye, J.; Long, R.; Antani, S.; Xue, Z.; and Huang, X. 2019. Selective Synthetic Augmentation with Quality Assurance. *arXiv preprint arXiv:1912.03837*.

Yang, Z.; Xu, X.; Zhang, J.; Wang, G.; Kalra, M. K.; and Yan, P. 2025. Chest X-Ray Foundation Model With Global and Local Representations Integration. *IEEE Transactions on Medical Imaging*, 44(12): 4787–4799.

Ye, J.; Xue, Y.; Long, L. R.; Antani, S.; Xue, Z.; Cheng, K.; and Huang, X. 2020. Synthetic Sample Selection via Reinforcement Learning. *arXiv preprint arXiv:2008.11331*.